Through the Looking Glass: Exploring Off-Axis Neutrino Oscillations with PRISM

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Many modern physics experiments rely on the creation of a neutrino beam. Because of the kinematics of its production, the energy spectrum of the produced neutrinos is dependent on the angle between the neutrino's momentum and the beam axis. These different neutrino spectra will then oscillate through flavor space as they travel from the accelerator to a detector. This study explores 3-neutrino oscillation for different off-axis angle regions, assuming the normal mass ordering, no CP-violation, and no matter effects. An analysis was conducted on the proportion of the total neutrino flux in each flavor state at given distances and off-axis angles.

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I. ACCELERATOR NEUTRINOS

A. How to produce a neutrino beam

Accelerator neutrino experiments are at the forefront of modern physics [1]. U.S.-based experiments like the Short Baseline Near Detector (SBND) use a neutrino beam created at Fermi National Accelerator Laboratory (FermiLab), and the future Deep Underground Neutrino Experiment (DUNE) will use another neutrino beam also produced at FermiLab. The first step in producing a beam of neutrinos is to collide high-energy protons with a target; FermiLab uses beryllium as the target for its Booster Neutrino Beam (BNB) with a proton beam energy of 8 GeV. This collision then produces many mesons, including $\pi^\pm$, π^0, and kaons. Mesons of the desired charge are then focused by a set of electromagnets, while mesons of the opposite charge are deflected by the same electromagnets. Next, as these mesons fly through space, they each decay into a charged lepton and a neutrino of the same flavor. The most common decays are $\pi^+ \to \mu^+ + \nu_\mu$ and $\pi^- \to \mu^- + \bar{\nu}_\mu$. The lepton is then stopped by a beam dump or the ground, while the neutrino passes through, creating a neutrino beam. An example schematic for a ν_μ beam production is shown in Figure 1.

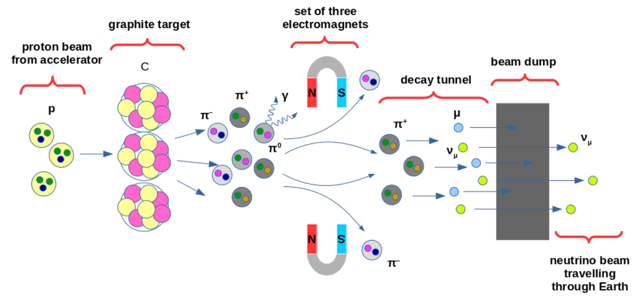

FIG. 1. A schematic showing the production process for a ν_μ beam [2]. In this example, graphene is used for the target.

B. Off-axis neutrino modeling

Without loss of generality, we will consider the production of a ν_μ beam from the decay of positively charged pions in flight ($\pi^+ \to \mu^+ + \nu_\mu$). During the neutrino beam production process, the electromagnets focus the pions to travel with momenta nearly parallel to the original proton beamline. However, in a pion's rest frame, its decay is isotropic. This means that in the laboratory frame, the neutrinos can be produced with non-zero momentum perpendicular to the beamline. The off-axis angle is the angle that the neutrino momentum makes with respect to the beamline, as shown in Figure 2.

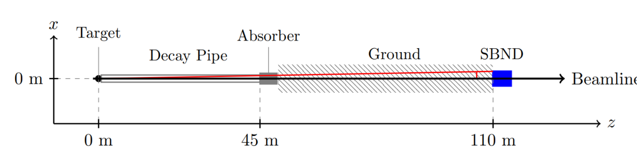

FIG. 2. Illustration of the BNB beamline and SBND at FermiLab. The origin at 0 m corresponds to the beginning of the target hit by the proton beam. The off-axis angle is that between the red line and the beamline [3].

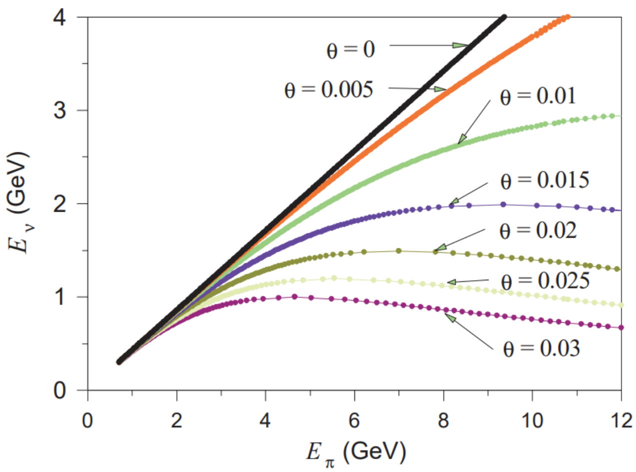

FIG. 3. Neutrino energy as a function of pion energy for different off-axis angles (in radians) [4].

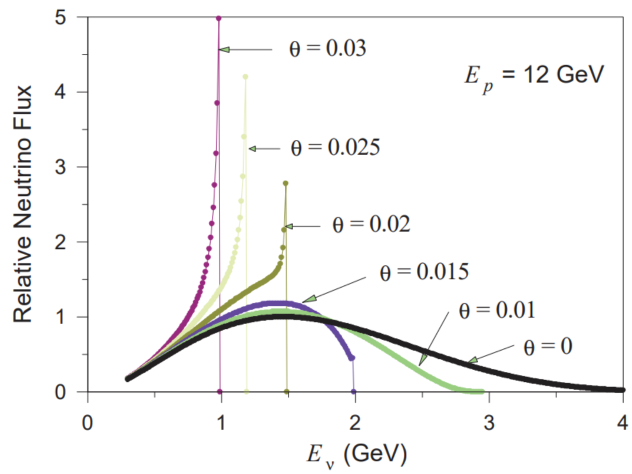

FIG. 4. Relative neutrino flux (in arbitrary units) as a function of the neutrino energy for different off-axis angles. A proton beam energy of E_p =12 GeV was used, and angles are in radians [4].

The neutrino spectrum as a function of the neutrino energy E_ν and proton beam energy E_p is given by:

$$\frac{d^2N}{dE_\nu d\cos\theta} \propto (E_p - E_\pi)^5 \frac{E_\pi E_\nu}{\sqrt{1 - \frac{4\pi^2 E_\nu^2}{(m_\pi^2 - m_\mu^2)^2}\left(\frac{1}{\cos^2\theta} - 1\right)}}. \tag{1}$$

where θ is the off-axis angle, m_π and m_μ are the masses of the charged pion and muon, respectively, and the pion energy E_π is given by:

$$E_\pi \approx \frac{2m_\pi^2 E_\nu}{(m_\pi^2 - m_\mu^2)\left(1 + \sqrt{1 - \frac{4\pi^2 E_\nu^2}{(m_\pi^2 - m_\mu^2)^2}\left(\frac{1}{\cos^2\theta} - 1\right)}\right)}. \tag{2}$$

Eqs. 1 and 2 come from [4], and a derivation of these equations from those presented in that source is in Appendix A.

Figure 3 shows the inverse of Eq. 2, the pion energy needed to produce a neutrino of energy E_ν at a given off-axis angle. The figure also shows that for a given off-axis angle, there is a maximum neutrino energy that can be produced. Also, a large range of pion energies contributes to neutrino energies near this peak, indicating that the neutrino flux would be greater near the maximum E_ν.

Figure 4 shows the flux defined in Eq. 1 versus the neutrino energy at different off-axis angles. This figure also demonstrates that a maximum neutrino energy exists for a given off-axis angle. This figure supports the idea that the neutrino fluxes are higher near this maximum neutrino energy. The figure also shows that the neutrino fluxes are more peaked at higher off-axis angles, albeit at lower neutrino energies.

C. PRISM

Since the different off-axis regions have different energy spectra, information can be gained by studying the different regions. This analysis technique is known as precision reaction independent spectrum measurement (PRISM). Looking at different off-axis angles can separate flux uncertainties and cross-section uncertainties [5]. This is because the flux has an angular dependence, but the neutrino-argon cross-section does not. This fact is used by the DUNE Collaboration to control uncertainties and reduce the model dependence of neutrino interactions [6].

Additionally, a preliminary study conducted by the SBND Collaboration found that π^0 production rates change as the off-axis angle changes [3]. At the Monte Carlo truth level, the π^0 rates are lower at higher off-axis angles, and so clever fiducialization of the detector can reduce the signal from π^0s. Since π^0s are a significant background to neutrino research [7], especially in searches for physics beyond the standard model, reducing this background can increase sensitivity to rare events.

Furthermore, experiments can take advantage of the different shapes of the off-axis spectra. The Tokai to Kamioka (T2K) experiment intentionally uses an off-axis neutrino beam: the Super-Kamiokande water Cherenkov detector is positioned 2.5° away from the center of the neutrino beam so that the neutrino energy spectrum is more peaked [8].

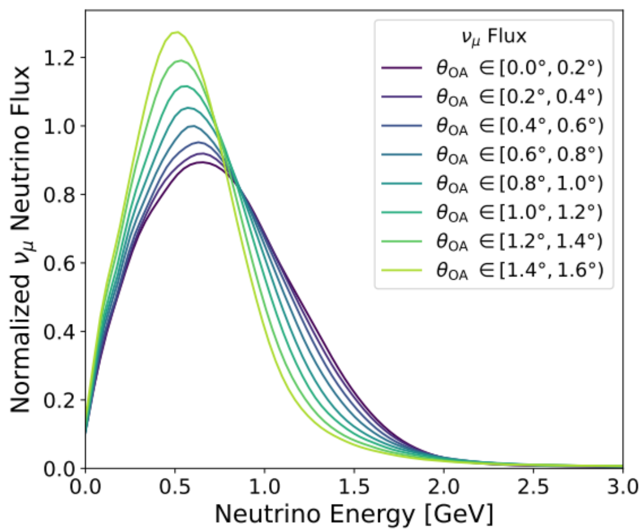

FIG. 5. Muon-neutrino fluxes at the front face of SBND for different off-axis angles, normalized so the area under each curve is unity [3].

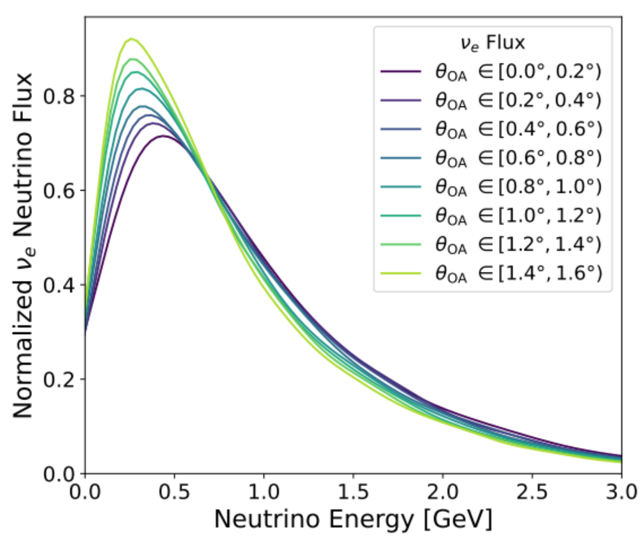

FIG. 6. Electron-neutrino fluxes at the front face of SBND for different off-axis angles, normalized so the area under each curve is unity [3].

D. Off-axis neutrino data

The model for off-axis neutrinos presented in [4] is, however, limited in its accuracy; the derivations for Eqs. 1 and 2 rely on several assumptions. This model assumes ultra-relativistic motion, where β for the pions is approximately 1. It assumes $E_\nu \gg m_\pi$, and so does not provide an accurate prediction when the energy of the emitted neutrino is comparable to the mass of the pion. This model makes a small-angle approximation when boosting the off-axis angle from the pion's rest frame to the laboratory frame. It also assumes that the pion decays immediately, rather than allowing for pions to travel some distance before decaying, which would spread out the neutrino spectrum. This also only considers one flavor of neutrino, whereas a real neutrino beam would also have ν_e and $\bar{\nu}_\mu$ from the secondary decay of the μ^+. Additionally, the true spectrum would also include contributions from the decay of other mesons produced in the proton-target collision, such as kaons.

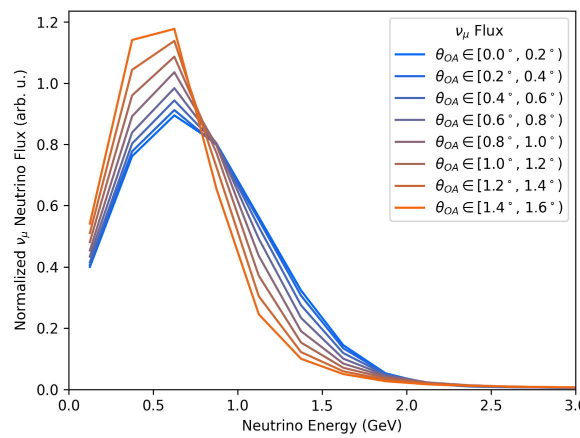

FIG. 7. Recreated muon-neutrino fluxes at the front face of SBND for different off-axis angles, normalized so the area under each curve is unity.

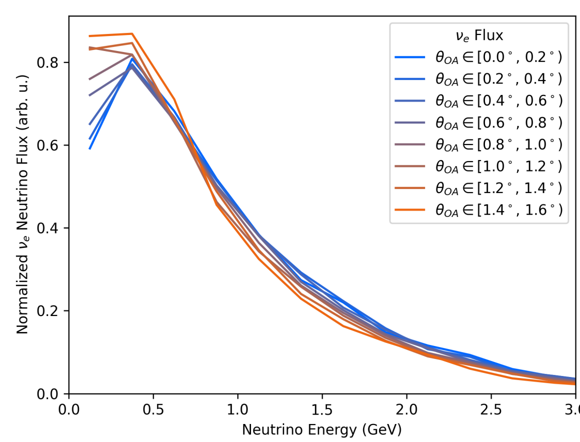

FIG. 8. Recreated electron-neutrino fluxes at the front face of SBND for different off-axis angles, normalized so the area under each curve is unity.

To more accurately model a beam of neutrinos, flux data was obtained from [3]. Figure 5 shows the ν_μ flux for different off-axis regions, each with an angular width of 0.2°. The underlying data is a series of histograms (one for each off-axis region) with bin widths of 0.25 GeV. The flux spectrum was smoothed using a kernel density estimation of the simulated neutrino energy distribution. The distributions were simulated using the specific Geant4 framework built by the MiniBooNE Collaboration [9], but then modified for SBND. The fluxes here are the neutrinos produced by the BNB and passing through the front face of SBND's LArTPC, situated 100 m downstream from the beryllium target. Similarly, Figure 6 shows the ν_e flux for different off-axis regions.

Figure 7 shows a recreation of the SBND data for the fluxes of ν_μ at different off-axis regions, but without smoothing. Similarly, Figure 8 shows a recreation of the SBND data for the fluxes of ν_e at different off-axis regions, also without smoothing. For ease of direct comparison to Figures 5 and 6, the midpoint of each histogram was plotted. It is clear from these plots that the ν_μ energy spectrum changes with the off-axis angle more drastically than the ν_e energy spectrum.

II. NEUTRINO OSCILLATIONS

This study examined the oscillations of the SBND neutrino energy spectra as they propagate through space. Table I contains the oscillation parameters used in this study, assuming the normal mass ordering, no CP-violations, and without Super Kamiokande atmospheric data. The Mikheyev-Smirnov-Wolfenstein effect for neutrinos traveling through matter was also not considered.

$$\begin{gathered}\theta_{12} = 33.82^\circ \\ \theta_{23} = 48.3^\circ \\ \theta_{13} = 8.61^\circ \\ \delta_{CP} = 0 \\ \Delta m^2_{21} = 7.39 \times 10^{-5}\ \text{eV}^2 \\ \Delta m^2_{32} = 2.449 \times 10^{-3}\ \text{eV}^2 \\ \Delta m^2_{31} = \Delta m^2_{32} + \Delta m^2_{21}\end{gathered}$$

TABLE I. Neutrino oscillation parameters used for this study, assuming the normal mass ordering, no CP-violations, and without Super Kamiokande atmospheric data. Values obtained from the Particle Data Group [10].

The Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrix for neutrino mixing is expressed as:

$$U = \begin{pmatrix} c_{12}\,c_{13} & s_{12}\,c_{13} & s_{13}\,e^{-i\delta_{\mathrm{CP}}} \\ -s_{12}\,c_{23} - c_{12}\,s_{13}\,s_{23}\,e^{i\delta_{\mathrm{CP}}} & c_{12}\,c_{23} - s_{12}\,s_{13}\,s_{23}\,e^{i\delta_{\mathrm{CP}}} & c_{13}\,s_{23} \\ s_{12}\,s_{23} - c_{12}\,s_{13}\,c_{23}\,e^{i\delta_{\mathrm{CP}}} & -c_{12}\,s_{23} - s_{12}\,s_{13}\,c_{23}\,e^{i\delta_{\mathrm{CP}}} & c_{13}\,c_{23} \end{pmatrix}$$

where $c_{ij} \equiv \cos\theta_{ij}$ and $s_{ij} \equiv \sin\theta_{ij}$. This form was taken from the Particle Data Group [10]. Notice that with no CP-violation, the exponentials evaluate to 1, so the matrix is entirely real. It was confirmed that when using the values from Table I, the magnitude of the PMNS matrix elements all fell within the 3σ confidence ranges of NuFIT 6.0 [11].

For a real mixing matrix, the probability of a neutrino oscillating from a flavor state α to a flavor state β is given by:

$$\mathcal{P}(\nu_\alpha \to \nu_\beta) = \delta_{\alpha\beta} - 4\sum_{i<j}^{n} U_{\alpha i}U_{\beta i}U_{\alpha j}U_{\beta j}\sin^2 X_{ij} \tag{3}$$

where

$$X_{ij} = \frac{(m_i^2 - m_j^2)L}{4E} = 1.267\Delta m^2_{ij}\frac{L}{E}. \tag{4}$$

Eqs. 3 and 4 were obtained from the Particle Data Group [10].

To calculate the neutrino energy spectrum of some flavor α at a distance L, the following formula was used:

$$\begin{aligned}\Phi_\alpha(L; E_\nu) &= \Phi_\alpha(L=0; E_\nu) \otimes \mathcal{P}(\nu_\alpha \to \nu_\alpha; L; E_\nu) \\ &\quad + \Phi_\beta(L=0; E_\nu) \otimes \mathcal{P}(\nu_\beta \to \nu_\alpha; L; E_\nu).\end{aligned} \tag{5}$$

The first line in Eq. 5 is the initial flux for neutrinos of the same flavor convolved with the survival probability. The second line is the initial flux for neutrinos of the other flavor convolved with the probability that it oscillates into the desired flavor. For a spectrum at an off-axis angle θ, the oscillation probability was calculated with distance $\frac{L}{\cos\theta}$.

Figure 9 shows the ν_μ energy spectra for different off-axis regions after traveling 1300 km from the front face of SBND. Similarly, Figure 10 shows the ν_e energy spectra at L =1300 km. A distance of 1300 km was chosen to reflect the distance that the DUNE far detector will be from FermiLab, although it should be noted that DUNE will use a new, higher-energy neutrino beam than the BNB used for SBND and this study. GIFs showing oscillations through intermediate distances can be found at the GitHub repository in [12]. It is highly recommended to watch those GIFs, as they show how the neutrino spectra evolve from the initial energy spectra in Figures 7 and 8 to the ones in Figures 9 and 10.

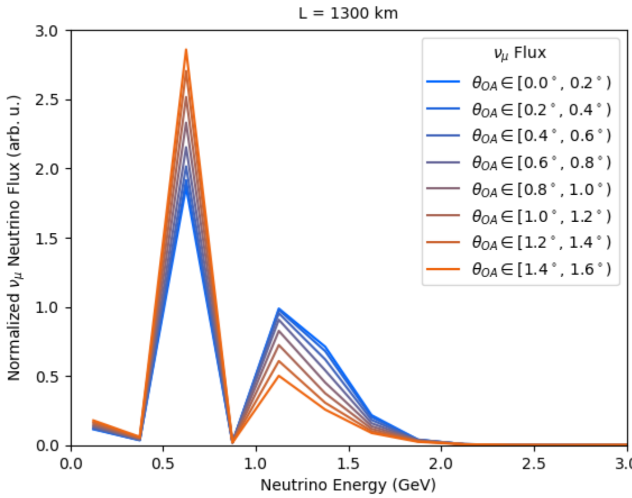

FIG. 9. ν_μ energy spectra at L =1300 km for different off-axis angles, normalized so the area under each curve is unity.

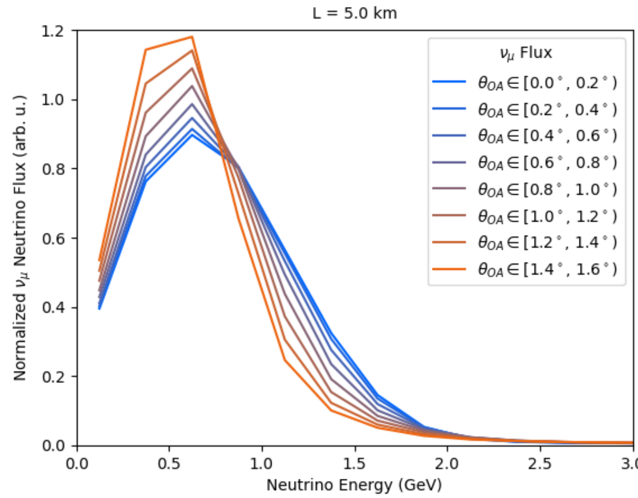

FIG. 11. ν_μ energy spectra at L =5 km for different off-axis angles, normalized so the area under each curve is unity.

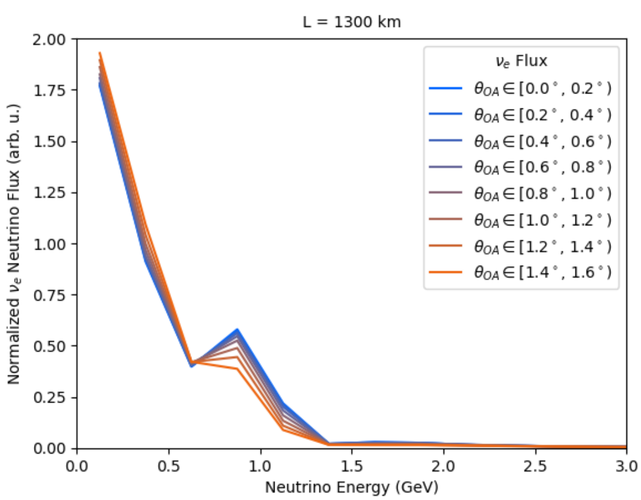

FIG. 10. ν_e energy spectra at L =1300 km for different off-axis angles, normalized so the area under each curve is unity.

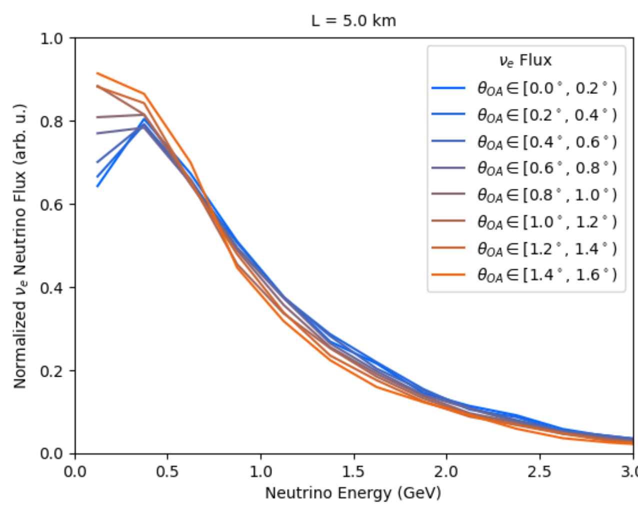

FIG. 12. ν_e energy spectra at L =5 km for different off-axis angles, normalized so the area under each curve is unity.

Figures 11 and 12 show the ν_μ and ν_e energy spectra at L =5 km, respectively. There is little to no change in the shape of these distributions, except for the very lowest energies in the ν_e energy spectra. This indicates that FermiLab's short baseline experiments (SBND to ICARUS) will not be able to see 3-neutrino oscillations, especially since that baseline difference is only 0.5 km.

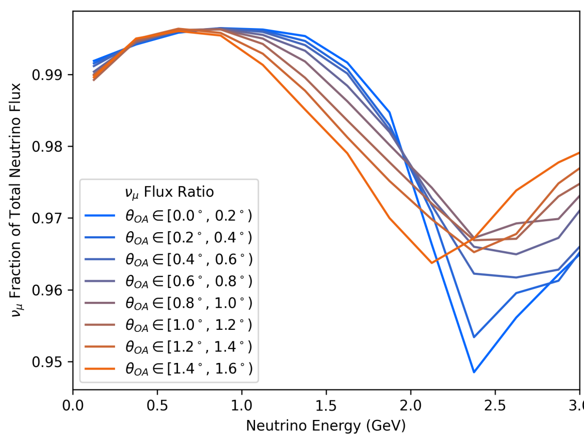

FIG. 13. Initial proportion of ν_μ of the total neutrino fluxes at L =0 km for different off-axis regions.

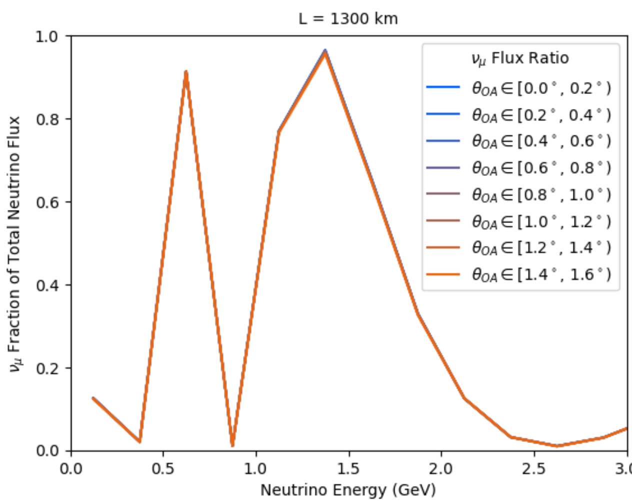

FIG. 14. Proportion of ν_μ of the total neutrino flux at L =1300 km for different off-axis regions.

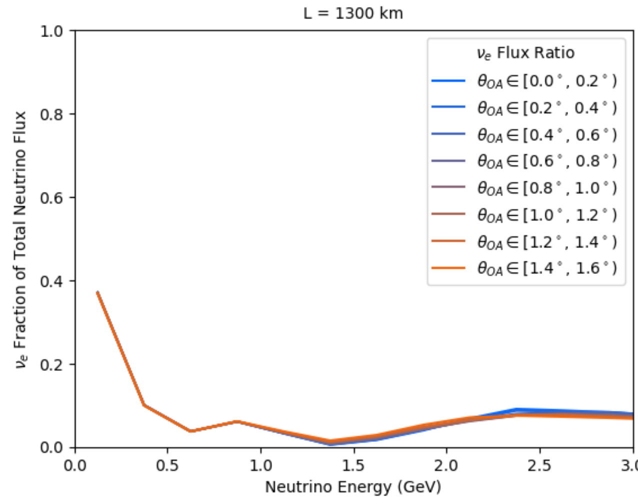

FIG. 15. Proportion of ν_e of the total neutrino flux at L =1300 km for different off-axis regions.

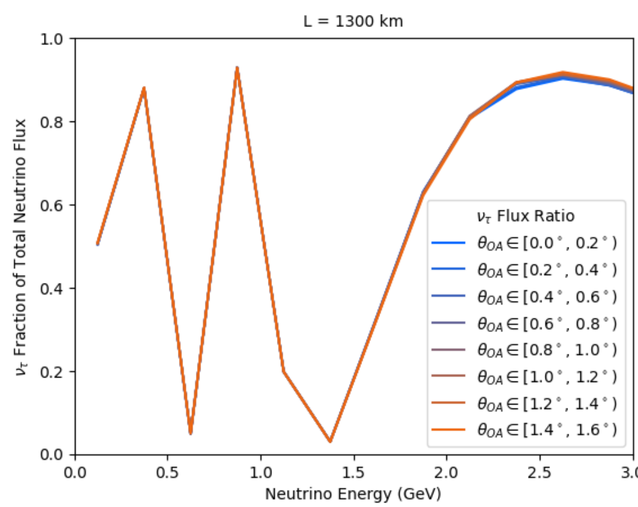

FIG. 16. Proportion of ν_τ of the total neutrino flux at L =1300 km for different off-axis regions.

III. PROPORTION OF ν_μ FLUX

An interesting pattern appears at the large distances: dips in the ν_μ energy spectra seem to correlate with peaks in the ν_e energy spectra and vice versa. The flux proportion of neutrinos with flavor $\alpha \in \{e, \mu, \tau\}$ is:

$$\frac{\Phi_\alpha(L; E_\nu)}{\Phi_e(L; E_\nu) + \Phi_\mu(L; E_\nu) + \Phi_\tau(L; E_\nu)} \tag{6}$$

Figure 13 shows the proportion of the total neutrino flux that is composed of ν_μ for each off-axis region. Initially, each off-axis region is dominated by ν_μs, with proportions greater than 0.95 over all energies.

Figures 14-16 show the proportion of the total neutrino flux which are in each flavor state. The broad trend from these figures is that the spectra for the different off-axis regions oscillate together, with no major separation in the fraction of the total flux for each flavor. Again, GIFs for the full evolution for each flavor state can be found in the GitHub repository [12].

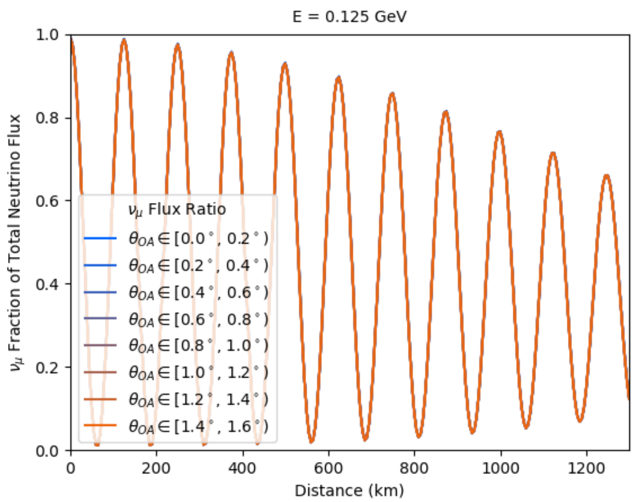

FIG. 17. Proportion of ν_μ of the total neutrino flux at E =0.125 GeV for different off-axis regions.

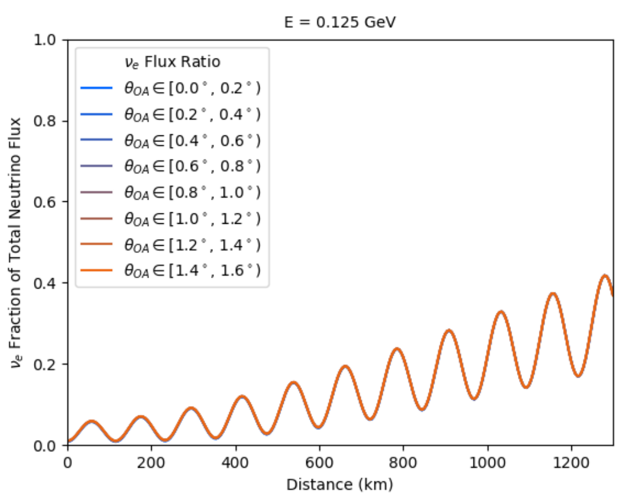

FIG. 18. Proportion of ν_e of the total neutrino flux at E =0.125 GeV for different off-axis regions.

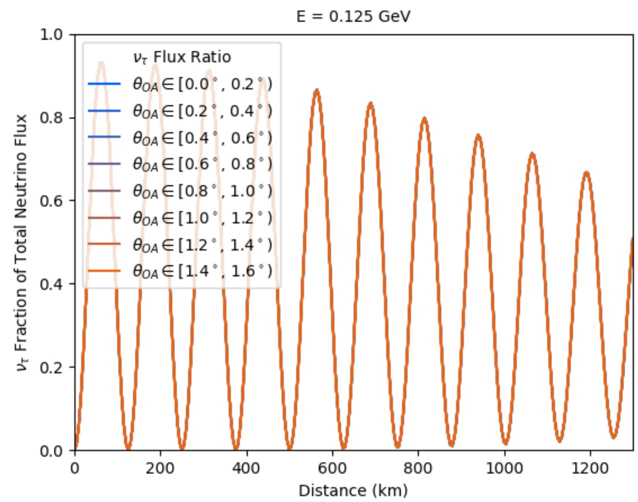

FIG. 19. Proportion of ν_τ of the total neutrino flux at E =0.125 GeV for different off-axis regions.

Figures 17-19 show the proportion of the total neutrino flux which are in each flavor state, but now with distance along the x-axis and with the energy constant at E_ν =0.125 GeV. Figures 20-22 show the proportions over distance but at a higher neutrino energy of E_ν =2.625 GeV. From these figures, it is clear to see the neutrino oscillations. At the low energies, the neutrinos primarily oscillate between ν_μ and ν_τ, with a small-but-getting-larger oscillatory contribution by the ν_e state. At the higher energy, the flux starts out almost entirely as ν_μs with some ν_es, but then oscillates almost completely to ν_τs at a distance of 1300 km and with slightly more ν_es. Again, there seems to be no significant difference between the off-axis regions. GIFs for these oscillations and stepping through different energy values can be found in the GitHub repository [12].

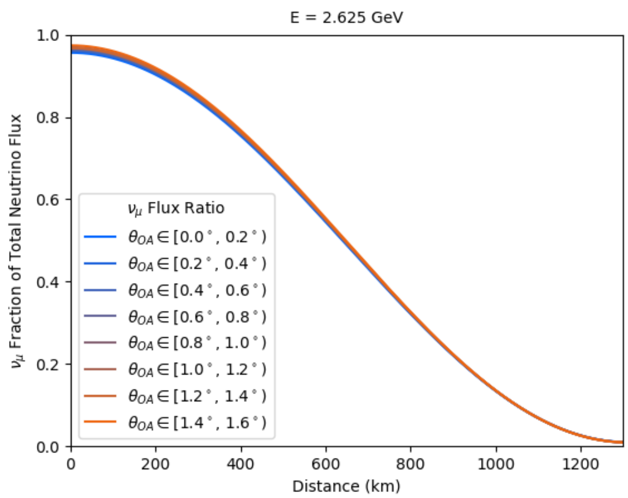

FIG. 20. Proportion of ν_μ of the total neutrino flux at $E =$2.625 GeV for different off-axis regions.

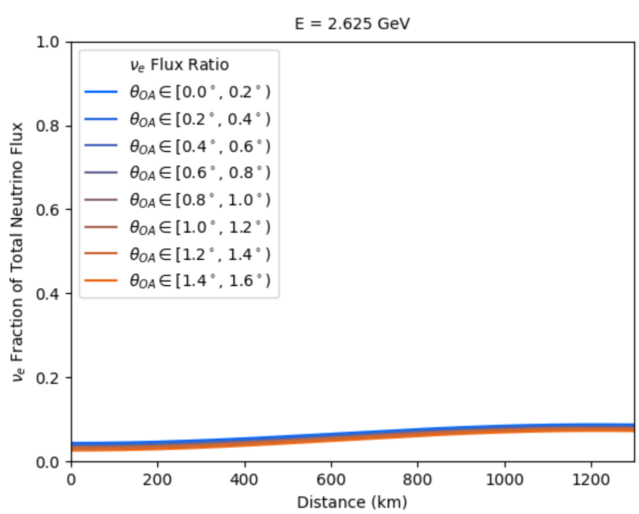

FIG. 21. Proportion of ν_e of the total neutrino flux at $E =$2.625 GeV for different off-axis regions.

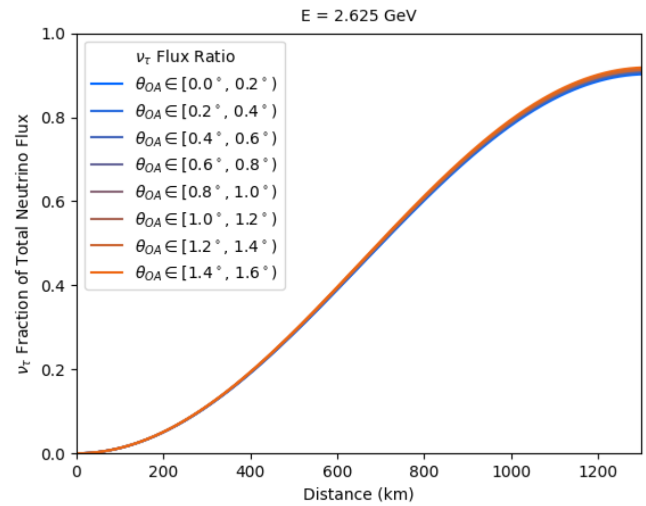

FIG. 22. Proportion of ν_τ of the total neutrino flux at $E =$2.625 GeV for different off-axis regions.

IV. FUTURE WORK

The physics in this study was done entirely at the Monte Carlo truth level and only considered the neutrino flux. However, the real observable for SBND and other LArTPC neutrino experiments is a convolution of the flux and the energy-dependent neutrino-argon cross-section. Additionally, the signal that would be detected would then be that convolved with the detector sensitivity and reconstruction (particle identification) efficiencies. A future study can look at how these different effects alter the off-axis neutrino spectra.

A necessary step to analyzing detector and reconstruction effects is to consider how the detector geometry might introduce artifacts in the signal. It may be that reconstruction efficiencies and detector sensitivity are dependent on the position of the neutrino event within the detector, and so somewhat correlated with the off-axis regions that pass through the detector. This correlation was preliminarily studied by the SBND Collaboration, but only at the Monte Carlo truth level and not at the reconstruction level. A future study can examine how the detector geometry affects the signal registered at different off-axis angles.

Lastly, it may be insightful to examine how the off-axis neutrino spectra would change when the oscillation parameters change. For instance, if the next study assumed an inverse mass ordering, a non-zero CP-violating phase, or using a 3+1 or 3+N neutrino model with ”sterile” flavors.

V. ACKNOWLEDGMENTS

The author would like to thank Dr. Georgia Karagiorgi for a wonderful semester of instruction about particle phenomenology. The author would also like to thank Dr. Mark Ross-Lonergan for mentorship and for introducing the topic of off-axis neutrinos and the PRISM technique. This study was conducted with data obtained from the SBND Collaboration.

Appendix A: Derivation of the off-axis neutrino spectrum equations

The following equations are provided in [4], with a * indicating that this quantity is in the pion's rest frame:

$$\frac{d^2N}{dE_\nu d\cos\theta} \propto (E_p - E_\pi)^5 \frac{E_\pi E_\nu}{\cos\theta^*}$$

$$\gamma_\pi = \frac{E_\pi}{m_\pi} = \frac{E_\nu}{E_\nu^*(1+\beta_\pi\cos\theta^*)} \approx \frac{E_\nu}{E_\nu^*(1+\cos\theta^*)}$$

$$\cos\theta^* \approx \sqrt{1 - \frac{E_\nu^2}{E_\nu^{*2}}\left(\frac{1}{\cos^2\theta} - 1\right)}$$

$$E_\nu^* = \frac{m_\pi^2 - m_\mu^2}{2m_\pi} = 29.8 \text{ MeV}$$

where the second equation makes the ultra-relativistic approximation and the third equation makes the small angle approximation.

The second equation can be rearranged to get

$$E_\pi \approx \frac{m_\pi E_\nu}{E_\nu^*(1+\cos\theta^*)}$$

Substituting the third and fourth equations into the first and second equations yields the flux and the pion energy in terms of quantities in the laboratory frame. This becomes exactly Eqs. 1 and 2.

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